VERIFICATION

FOR RIP ROUTER:

```
*Mar 21, 13:44:10.444: %SYS-5-CONFIG_I: Configured from console by console
R1_RIP#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.10.0/30 is directly connected, GigabitEthernet0/0/1
L       10.10.10.1/32 is directly connected, GigabitEthernet0/0/1
     172.16.0.0/24 is subnetted, 3 subnets
D       172.16.10.0 [90/28416] via 10.10.10.2, 00:25:04, GigabitEthernet0/0/1
D       172.16.20.0 [90/28416] via 10.10.10.2, 00:25:04, GigabitEthernet0/0/1
D       172.16.30.0 [90/28416] via 10.10.10.2, 00:25:04, GigabitEthernet0/0/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0/0.20
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0/0.30

R1_RIP#
```

```
R1_RIP#show ip protocols

Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
  EIGRP maximum hopcount 100
  EIGRP maximum metric variance 1
Redistributing: eigrp 100, rip
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
     10.10.10.0/30
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.10.10.2      90            727497
  Distance: internal 90 external 170

Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 22 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip, eigrp 100
Default version control: send version 2, receive 2
  Interface             Send  Recv  Triggered RIP  Key-chain
  GigabitEthernet0/0/1  22
  GigabitEthernet0/0/0.1022
  GigabitEthernet0/0/0.2022
  GigabitEthernet0/0/0.3022
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
            10.0.0.0
            192.168.10.0
            192.168.20.0
            192.168.30.0
Passive Interface(s):
Routing Information Sources:
            Gateway         Distance      Last Update
Distance: (default is 120)
```

```
R1_RIP#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface      Hold Uptime    SRTT   RTO   Q   Seq
                                   (sec)          (ms)        Cnt  Num
0   10.10.10.2      Gig0/0/1       14   00:27:07  40     1000  0   4

R1_RIP#
```

FOR EIGRP ROUTER:

```
R2_EIGRP>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.10.0/30 is directly connected, GigabitEthernet0/0/1
L       10.10.10.2/32 is directly connected, GigabitEthernet0/0/1
     172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C       172.16.10.0/24 is directly connected, GigabitEthernet0/0/0.10
L       172.16.10.1/32 is directly connected, GigabitEthernet0/0/0.10
C       172.16.20.0/24 is directly connected, GigabitEthernet0/0/0.20
L       172.16.20.1/32 is directly connected, GigabitEthernet0/0/0.20
C       172.16.30.0/24 is directly connected, GigabitEthernet0/0/0.30
L       172.16.30.1/32 is directly connected, GigabitEthernet0/0/0.30
R    192.168.10.0/24 [120/1] via 10.10.10.1, 00:00:04, GigabitEthernet0/0/1
R    192.168.20.0/24 [120/1] via 10.10.10.1, 00:00:04, GigabitEthernet0/0/1
R    192.168.30.0/24 [120/1] via 10.10.10.1, 00:00:04, GigabitEthernet0/0/1
```

```
R2_EIGRP>
R2_EIGRP>show ip protocols

Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
  EIGRP maximum hopcount 100
  EIGRP maximum metric variance 1
Redistributing: eigrp 100, rip
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
     172.16.10.0/24
     172.16.20.0/24
     172.16.30.0/24
     10.10.10.0/30
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.10.10.1      90            676673
  Distance: internal 90 external 170

Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 7 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip, eigrp 100
Default version control: send version 2, receive 2
  Interface             Send  Recv  Triggered RIP  Key-chain
  GigabitEthernet0/0/1  22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
            10.0.0.0
Passive Interface(s):
Routing Information Sources:
            Gateway         Distance      Last Update
            10.10.10.1           120      00:00:18
Distance: (default is 120)
R2_EIGRP>
```

```
R2_EIGRP>show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface      Hold Uptime    SRTT   RTO   Q   Seq
                                   (sec)          (ms)        Cnt  Num
0   10.10.10.1      Gig0/0/1       13   00:32:17  40     1000  0   13
```

VERIFICATION ON SWITCHES:

SW_RIP:

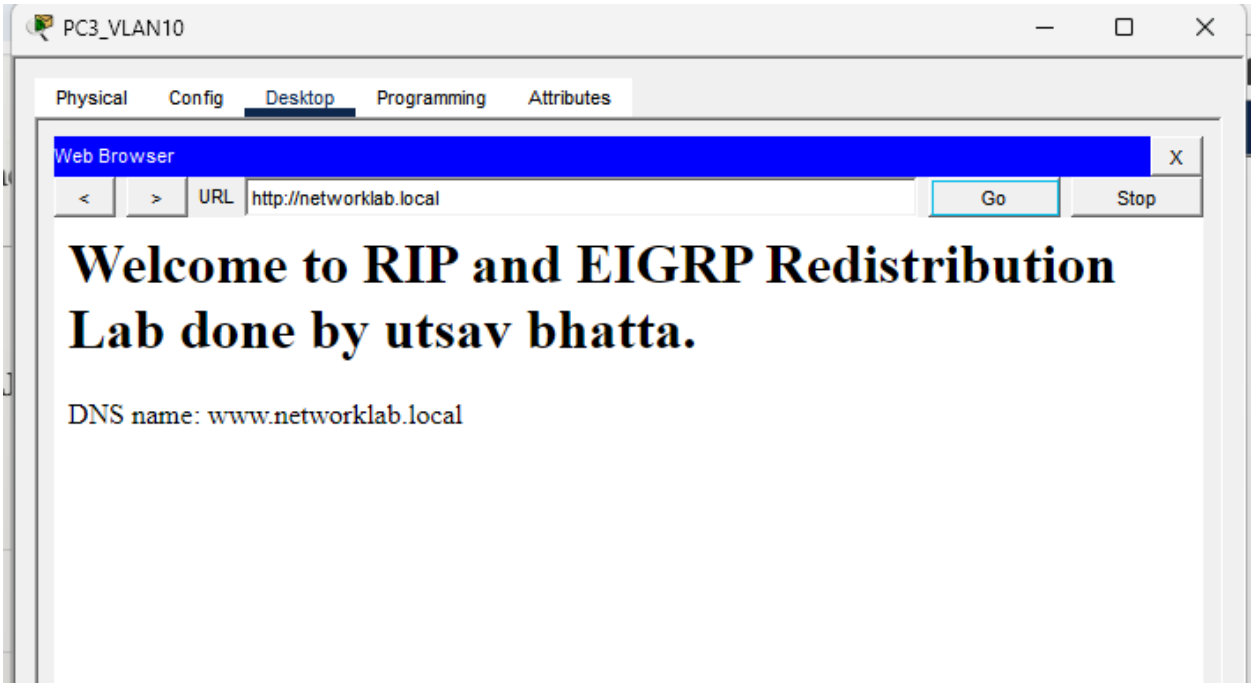
```
SW1_RIP>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   VLAN10_USERS                     active    Fa0/3
20   VLAN20_USERS                     active    Fa0/4
30   VLAN30_SERVERS                   active    Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1_RIP>
```

```
SW1_RIP>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30
```

SW_EIGRP:

```
SW2_EIGRP>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   VLAN10_USERS                     active    Fa0/5
20   VLAN20_USERS                     active    Fa0/6
30   VLAN30_SERVERS                   active    Fa0/2, Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW2_EIGRP>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30
```

PING TEST IN PC1_VLAN10:

```
C:\>ping 172.16.30.30

Pinging 172.16.30.30 with 32 bytes of data:

Reply from 172.16.30.30: bytes=32 time<1ms TTL=126
Reply from 172.16.30.30: bytes=32 time<1ms TTL=126
Reply from 172.16.30.30: bytes=32 time<1ms TTL=126
Reply from 172.16.30.30: bytes=32 time<1ms TTL=126

Ping statistics for 172.16.30.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Browser test in PC3_VLAN10: